

Lecture 2

Linear Independence, Basis, Dimension

- Linear Independence
- Basis and Dimension
- Connections with Artificial Intelligence

Contents

- Linear Independence
- Basis and Dimension
- Connections with Artificial Intelligence

Linear Independence

- Whether the homogeneous system $A\mathbf{x} = 0$ has a unique solution or many solutions is an important question.
- The question is equivalent to whether there exists $x_1, \dots, x_n$, not all zero, such that

$$\sum_{i=1}^n x_i \mathbf{a}_i$$
$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{0}.$$

Linear Independence

- The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in $\mathbb{R}^m$ are said to be **linearly independent** if

$$\langle c, v \rangle, c, v \in \mathbb{R}^m$$

$$\underline{c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}}$$

implies that all the scalars $c_1, \dots, c_n$ are 0.

- The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in $\mathbb{R}^m$ are said to be **linearly dependent** if there exists scalars $c_1, c_2, \dots, c_n$, not all zero, such that

$$\underline{c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}} \quad c \neq 0$$

Exercise

Determine whether the following sets of vectors are linearly dependent or not.

$$v_2 = 2v_1$$

def

$v_1, \dots, v_n$ are linearly ind. if

$$c_1 v_1 + \dots + c_n v_n = 0$$

implies $c_1 = \dots = c_n = 0$.

def

dependent: $\exists c \neq 0$ s.t.

$$c^T v = 0$$

$$\textcircled{1} c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = 0$$

$$c_1 = -2$$

$$c_2 = 1$$

1. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right\}$

2. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 0 \end{bmatrix} \right\}$

3. $\{0\}$ v_1, v_2 $v_3 = -5v_1 + 7v_2$

HW 4. $\{v_1, v_2, 0\}$

e.g. assume v_1, v_2 are independent

Linear Independence, Basis, Dimension

$\therefore$ linearly dependent

$$\textcircled{2} \begin{matrix} \downarrow -5 & \downarrow 7 & \downarrow -1 \end{matrix} c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 7 \\ 0 \end{bmatrix} = 0$$

Linear Independence and System of Linear Equations

- To determine whether vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly dependent or not, we can check whether the system $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n = A\mathbf{x} = \mathbf{0}$ has a non-trivial solution or not.
- In other words, if the columns of A are linearly independent, the system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. If the columns of A are linearly dependent, the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
 - For a square matrix A , its columns are linearly dependent if and only if A is singular.
 - For an $m \times n$ matrix A with $m < n$, its columns are linearly dependent.

Linear Independence and System of Linear Equations

- To determine whether vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly dependent or not, we can check whether the system $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n = A\mathbf{x} = \mathbf{0}$ has a non-trivial solution or not.
- In other words, if the columns of A are linearly independent, the system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. If the columns of A are linearly dependent, the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
- For a square matrix A , its columns are linearly dependent if and only if A is singular. (no inverse)
- For an $m \times n$ matrix A with $m < n$, its columns are linearly dependent.

$\text{Null}(A) = \emptyset$

$$\underbrace{m}_{\text{\# of features}} \times \underbrace{n}_{\text{\# of training samples}} \left. \vphantom{\begin{matrix} m \\ n \end{matrix}} \right\} \begin{matrix} \text{\# constraints} < \text{\# variables} \end{matrix}$$

Vector Space

A set $\mathcal{V}$, on which two operations addition and scalar multiplication are defined, is a **vector space** if the following axioms are satisfied:

A1. $x + y = y + x$ for any $x, y \in \mathcal{V}$.

symmetry

A2. $(x + y) + z = x + (y + z)$ for any $x, y, z \in \mathcal{V}$.

associative

A3. There exists $\mathbf{0} \in \mathcal{V}$ such that $x + \mathbf{0} = x$ for each $x \in \mathcal{V}$.

0 is a additive identity element

A4. For each $x \in \mathcal{V}$, there exists $x' \in \mathcal{V}$ such that $x + x' = \mathbf{0}$, where x' is usually denoted as $-x$.

existence of additive inverse

A5. $\alpha(x + y) = \alpha x + \alpha y$ for each scalar α and any $x, y \in \mathcal{V}$.

$\alpha \in \mathbb{R}$

distributive

A6. $(\alpha + \beta)x = \alpha x + \beta x$ for any scalars α and β and any $x \in \mathcal{V}$.

A7. $(\alpha\beta)x = \alpha(\beta x)$ for any scalars α and β and any $x \in \mathcal{V}$.

assoc.

A8. $1x = x$ for all $x \in \mathcal{V}$.

multiplicative identity

Examples of Vector Space

- $\mathbb{R}^n, n \geq 1$

$$n=2.$$
$$f(x) = \underline{C_0} + \underline{C_1}x + \underline{C_2}x^2 + \underline{C_3}x^3 + \underline{C_4}x^4 + \underline{C_5}x^5 + \underline{C_6}x^6$$
$$[C_0 \ C_1 \ \cdots \ C_5]$$

- $\mathbb{R}^{m \times n}$

- Let P_n denote the set of all polynomials of degree less than n .
- Let $C[a, b]$ denote the set of all real-valued functions that are defined and continuous on $[a, b]$.

↳ Reduced Kernel Hilbert Space
(RKHS)

Operations on General Vector Space

Linear combination, linear span and linear independence can be defined on general vector space $\mathcal{V}$:

- **linear combination:** $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n \in \mathcal{V}$. $c_i \in \mathbb{R}$
- **linear span:** $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \{\sum_i c_i \mathbf{v}_i : c_i \in \mathbb{R}\} \subset \mathcal{V}$.
- **linear independence:** $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$ implies $c_1, \dots, c_n$ are all zero.

Example

Reduced echelon form. $\rightarrow$ Find independent basis
in a matrix.

$\rightarrow$ solve linear system

- How to test matrices $M_1, \dots, M_k \in \mathbb{R}^{m \times n}$ are linearly independent?

- Are the matrices $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ linearly independent?

Example

- How to test vectors (polynomials) $p_1, p_2, \dots, p_k$ are linearly independent in P_n ?
- Are the polynomials

$$p_1(x) = x^2 + 3, \quad p_2(x) = 2x^2 + x, \quad p_3(x) = 8x + 7$$

in P_3 linearly independent?

Linear Independence

- How to test vectors (functions) $f_1, \dots, f_k$ are linearly independent in $C[a, b]$?
- Are the functions $x, x^2, \sin(x) \in C[-2, 2]$ linearly independent?

Linear Independence

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space $\mathcal{V}$ are linearly dependent if and only if for a certain $k \in \{1, 2, \dots, n\}$, $\mathbf{v}_k$ is a linear combination of the other vectors.

Minimum Spanning Set

- For a vector space $\mathcal{V}$, we call $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset \mathcal{V}$ a **spanning set of $\mathcal{V}$** if $\text{Span}(\mathcal{S}) = \mathcal{V}$.
- For a vector space $\mathcal{V}$, we say $\mathcal{S} \subset \mathcal{V}$ is a **minimal spanning set of $\mathcal{V}$** if $\mathcal{V}$ cannot be generated by any proper subset of $\mathcal{S}$.
- Suppose $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a minimal spanning set of a vector space $\mathcal{V}$. Then $\mathcal{S}$ is linearly independent.

Minimum Spanning Set

- For a vector space $\mathcal{V}$, we call $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset \mathcal{V}$ a **spanning set of $\mathcal{V}$** if $\text{Span}(\mathcal{S}) = \mathcal{V}$.
- For a vector space $\mathcal{V}$, we say $\mathcal{S} \subset \mathcal{V}$ is a **minimal spanning set of $\mathcal{V}$** if $\mathcal{V}$ cannot be generated by any proper subset of $\mathcal{S}$.
- Suppose $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a minimal spanning set of a vector space $\mathcal{V}$. Then $\mathcal{S}$ is linearly independent.

Minimum Spanning Set

- For a vector space $\mathcal{V}$, we call $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset \mathcal{V}$ a **spanning set of $\mathcal{V}$** if $\text{Span}(\mathcal{S}) = \mathcal{V}$.
- For a vector space $\mathcal{V}$, we say $\mathcal{S} \subset \mathcal{V}$ is a **minimal spanning set of $\mathcal{V}$** if $\mathcal{V}$ cannot be generated by any proper subset of $\mathcal{S}$.
- Suppose $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a minimal spanning set of a vector space $\mathcal{V}$. Then $\mathcal{S}$ is linearly independent.

Contents

- Linear Independence
- Basis and Dimension
- Connections with Artificial Intelligence

Basis

The vectors $v_1, v_2, \dots, v_n$ form a **basis** for a vector space $\mathcal{V}$ if

1. $v_1, v_2, \dots, v_n$ are linearly independent, and
2. $v_1, v_2, \dots, v_n$ span $\mathcal{V}$.

How to determine whether a set $\mathcal{B}$ of vectors form a basis of a vector space $\mathcal{V}$?

- First, check that $\mathcal{B}$ is a subset of $\mathcal{V}$.
- Second, verify that $\mathcal{B}$ is linearly independent.
- Third, verify that for any $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} \in \text{Span}\{\mathcal{B}\}$.

Basis

The vectors $v_1, v_2, \dots, v_n$ form a **basis** for a vector space $\mathcal{V}$ if

1. $v_1, v_2, \dots, v_n$ are linearly independent, and
2. $v_1, v_2, \dots, v_n$ span $\mathcal{V}$.

How to determine whether a set $\mathcal{B}$ of vectors form a basis of a vector space $\mathcal{V}$?

- First, check that $\mathcal{B}$ is a subset of $\mathcal{V}$.
- Second, verify that $\mathcal{B}$ is linearly independent.
- Third, verify that for any $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} \in \text{Span}\{\mathcal{B}\}$.

Example

Are the following sets a basis for $\mathbb{R}^2$ or not?

- $\mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

- $\mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$

- $\mathcal{B}_3 = \left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$

- $\mathcal{B}_4 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$

- $\mathcal{B}_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

Example

Are the following sets a basis for $\mathbb{R}^2$ or not?

$$\bullet \mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_3 = \left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_4 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Example

Are the following sets a basis for $\mathbb{R}^2$ or not?

$$\bullet \mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_3 = \left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_4 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Example

Are the following sets a basis for $\mathbb{R}^2$ or not?

$$\bullet \mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_3 = \left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_4 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Example

Are the following sets a basis for $\mathbb{R}^2$ or not?

$$\bullet \mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_3 = \left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_4 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$$

$$\bullet \mathcal{B}_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Example

Given a vector space $\mathbb{R}^{2 \times 2}$, the set $\mathcal{B}$ consisting of

$$B_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, B_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

is a basis of $\mathbb{R}^{2 \times 2}$.

Example

Are the polynomials

$$p_1(x) = x^2 + 3, \quad p_2(x) = 2x^2 + x, \quad p_3(x) = 8x + 7$$

form a basis of P_3 ?

Dimension

- If a vector space $\mathcal{V}$ has a basis consisting of n vectors, we say that $\mathcal{V}$ has **dimension** n .
- The subspace $\{\mathbf{0}\}$ of $\mathcal{V}$ is said to have dimension 0.
- $\mathcal{V}$ is said to be *finite dimensional* if there is a finite set of vectors that spans $\mathcal{V}$; otherwise, we say that $\mathcal{V}$ is *infinite dimensional*.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n+1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite-dimensional.
- $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n+1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite-dimensional.
- $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n+1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite-dimensional.
- $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^n\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n+1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite-dimensional.
- $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^n\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n+1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite-dimensional.
- $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
 - $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
 - $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
 - $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
 - $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
 - $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
 - $C[a, b]$ is infinite dimensional.

Example

- What is the dimension of $\mathbb{R}^n$?
 - The standard basis has n vectors.
- What is the dimension of $\mathbb{R}^{m \times n}$?
 - All the $m \times n$ matrices with only one non-zero entry 1 form a basis.
- What is the dimension of P_n ?
 - $\{1, x, x^2, \dots, x^{n-1}\}$ forms a basis of P_n .
- Let P be the vector space of all polynomials.
 - If P has a finite dimension n , then any $n + 1$ polynomials would be linearly dependent. Find a contradiction.
 - P is infinite dimensional.
- $C[a, b]$ is infinite dimensional.

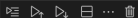
Contents

- Linear Independence
- Basis and Dimension
- Connections with Artificial Intelligence

Regression with Linear Dependent Data

1. Linear Dependence in Features

If two features are linearly dependent, they carry redundant information, which does not improve a model.



```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Create data
np.random.seed(0)
X1 = np.random.rand(100, 1) # independent feature
X2 = np.random.randn(100, 1) # independent feature
X3 = 2 * X1 + 3 * X2         # dependent feature (perfectly correlated)
y = 3 * X1.squeeze() + 6 * X2.squeeze() + np.random.randn(100) * 0.1

# Train models
reg1 = LinearRegression().fit(np.hstack([X1, X2]), y) # with one feature
reg2 = LinearRegression().fit(np.hstack([X1, X2, X3]), y) # with redundant feature

print("R^2 with two features:", reg1.score(np.hstack([X1, X2]), y))
print("R^2 with redundant features (full):", reg2.score(np.hstack([X1, X2, X3]), y))
```

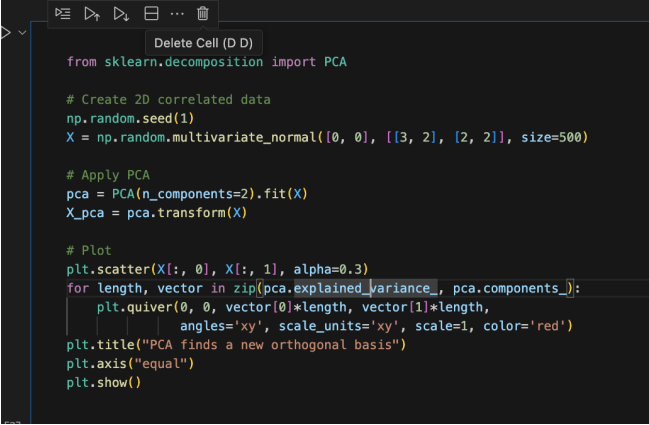
R^2 with two features: 0.999740798192029

R^2 with redundant features (full): 0.999740798192029

AI for Finding Basis

2. Basis & PCA for Dimensionality Reduction

PCA finds a new orthogonal basis that captures maximum variance in the data.

A screenshot of a Jupyter Notebook interface. At the top, there is a toolbar with icons for running, stepping through, and deleting cells. Below the toolbar, a button labeled "Delete Cell (D D)" is visible. The main area contains Python code for PCA. The code imports PCA from sklearn.decomposition, creates a 2D correlated dataset X, applies PCA with 2 components, and plots the original data points as semi-transparent blue dots. It also plots the principal components as red arrows originating from the center, with the first component (PC1) being longer than the second (PC2). The plot is titled "PCA finds a new orthogonal basis".

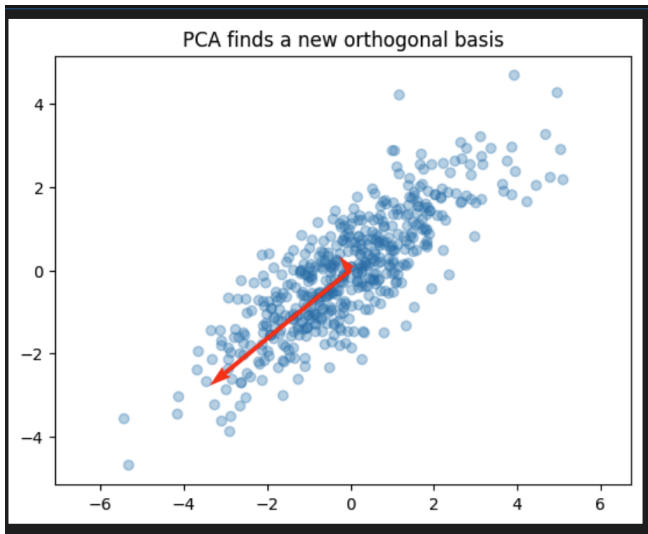
```
from sklearn.decomposition import PCA

# Create 2D correlated data
np.random.seed(1)
X = np.random.multivariate_normal([0, 0], [[3, 2], [2, 2]], size=500)

# Apply PCA
pca = PCA(n_components=2).fit(X)
X_pca = pca.transform(X)

# Plot
plt.scatter(X[:, 0], X[:, 1], alpha=0.3)
for length, vector in zip(pca.explained_variance_, pca.components_):
    plt.quiver(0, 0, vector[0]*length, vector[1]*length,
               angles='xy', scale_units='xy', scale=1, color='red')
plt.title("PCA finds a new orthogonal basis")
plt.axis("equal")
plt.show()
```

AI for Finding Basis



Infinite-Dimension AI Model

3. Infinite-Dimensional Spaces: Kernel Trick

Kernel methods (e.g., SVM with RBF kernel) operate in infinite-dimensional spaces implicitly, while computations remain finite.

```
from sklearn.svm import SVC
from sklearn.datasets import make_moons

# Generate dataset
X, y = make_moons(n_samples=200, noise=0.1, random_state=0)

# Train SVM with RBF kernel (infinite-dimensional space)
clf = SVC(kernel='rbf', C=10).fit(X, y)

# Plot decision boundary
xx, yy = np.meshgrid(np.linspace(-2, 3, 200), np.linspace(-1.5, 2, 200))
Z = clf.predict(np.c_[xx.ravel(), yy.ravel()]).reshape(xx.shape)

plt.contourf(xx, yy, Z, alpha=0.3)
plt.scatter(X[:, 0], X[:, 1], c=y, edgecolors='k')
plt.title("SVM with RBF kernel (infinite-dimensional space)")
plt.show()
```

Recommended Reading:

- Two-sample Test with Kernel Projected Wasserstein Distance
- Statistical and Computational Guarantees of Kernel Max-Sliced Wasserstein Distances
- Variable Selection for Kernel Two-Sample Tests

Infinite-Dimension AI Model

